Verification-Aware RISC-V SoC + TPU

PROJECT BRIEF | CONFERENCE / INTERVIEW

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A system-level study of accelerator performance and verification reachability; the contribution is the evaluation methodology, not a new TPU architecture.

SELECTED SIGMOID WORKLOAD

3.61×

1,343 cycles pre-TPU to 372 post-integration cycles. TPU transaction is separate: 67 cycles (7 TX + 58 compute + 2 RX).

ACTIVE UVM MUTATION RESULT

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Eight viable mutants detected; two of ten were equivalent. Directed firmware and passive UVM each detected 2 of 8 viable mutants.

MERGED UVM COVERAGE

43.65%

26 of 109 bins covered. The flow separately reports a 23.85% covered-bin hit rate; both metrics are retained as reported.

SYSTEM AND VERIFICATION SETUP

- **SoC:** PicoRV32 RV32I processor, native valid/ready bus, and memory-mapped peripheral windows.
- **TPU integration:** 32-bit MMIO wrapper exposes paired accesses to 64-bit weights, inputs, and results; stream master sends seven input beats and captures two result beats.
- **UVM modes:** Passive UVM observes firmware traffic; active UVM holds the CPU in reset and drives the native bus directly. RAL provides frontdoor register access.
- **Numerical checking:** A C DPI reference model supplies expected TPU results to the UVM scoreboard.

TOOLS USED

QuestaSim, SystemVerilog/UVM/RAL, C DPI, SymbiYosys with Yosys-SMTBMC and Z3, and QuestaSim RNM.

FORMAL RESULTS (BOUNDED)

- **CPU/RAM:** Abstract bus-property group recorded PASS at depth 10.
- **TPU control:** Five assertions passed in 30 assertion-depth checks across bounds 1, 2, 4, 6, 8, and 10. Both cover goals were reached by bound 6; witness traces show axis_start at step 3 and m_axis_tvalid at step 4.
- **Formal mutation checks:** Five targeted AXI-Stream/control mutants produced counterexamples under their paired properties. This is a separate mutant set from the UVM campaign.

SENSOR RNM MODEL

84 reported scenarios: 42 noise-off and 42 seeded-noise; no alarm mismatches. Temperature-error MAE rose from 0.0714 °C to 0.4524 °C. Input-temperature/readback correlation was 1.000000 and 0.999997, respectively.

SCOPE TO STATE CLEARLY

The 372-cycle benchmark and 67-cycle accelerator transaction measure different scopes. Formal results are bounded; sensor, interconnect, and TPU-interface formal targets are not established by the saved runs. RNM metrics describe the implemented model, not a physical sensor. Coverage percentages are distinct flow metrics.

CORE TAKEAWAY

In the tested campaign, direct native-bus stimulus exposed viable RTL faults that the selected firmware workload did not reach; formal and RNM added bounded and model-level checks.

Basis: Phase 7 mutation summary and merged coverage report; Phase 8 formal result records; reported performance and RNM results. Results are limited to the stated tests and models.